

FoodResQ

Turn Surplus into Impact

Surplus Food Rescue and Last-Mile Coordination Platform

FeedBox College Club Hackathon
Problem Statement 2 — The Good Food Gets Wasted

Team Zero/Zero Project Report

1. Introduction :

Food waste is a common problem, especially when restaurants, hostels, households and events have more food than they can use. At the same time, nearby organizations may need food. The problem is often not only finding surplus food, but also making sure it reaches the right place at the right time. FoodResQ is developed as a simple platform to coordinate this process. It connects food donors, NGOs and volunteers so that surplus food can be listed, claimed and delivered in a more organized way. The main idea is simple:

Donor → NGO → Volunteer → Delivery → Verification

2. Problem Statement :

The selected problem statement focuses on edible surplus food being wasted while suitable recipients may be nearby. It highlights the need for surplus food reporting, nearby availability, pickup coordination and timely communication.

The main problems we identified are:

- Surplus food may only remain usable for a limited period.
 - People and organizations may not know where surplus food is available.
 - Even when an NGO is found, transportation can become a problem.
 - There is limited visibility into how much food has actually been redistributed.
 - People who regularly contribute do not have a simple way to build community recognition.
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3. Proposed Solution :

FoodResQ provides three different interfaces for the three main participants in the process.

- Food Donor A restaurant, hostel, household or other donor can create a surplus-food listing.
- The donor provides details such as food type, quantity, pickup location, deadline and delivery contribution.
- NGO / Recipient An NGO can see available food and choose a suitable listing.
- Once the NGO claims the food, a delivery request is created.

- Volunteer A volunteer can see available delivery requests, accept one, pick up the food and mark it as delivered.
 - Verification After delivery, the NGO confirms that the food was received and gives a rating based on quality, quantity and timeliness.
 - This information is then used for the community impact scores.
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4. How the System Works :

The main workflow of FoodResQ is:

1. A donor posts surplus food.
2. The food becomes available to nearby recipients.
3. An NGO claims the food.
4. A delivery request becomes available to volunteers.
5. A volunteer accepts the delivery.
6. The volunteer picks up the food and marks the pickup.
7. The volunteer delivers the food and marks it as delivered.
8. The NGO confirms receipt.
9. The NGO reviews the quality, quantity and timeliness.

10. The successful transaction contributes to the donor and volunteer impact scores.

The delivery status used in the prototype is :

Available → Claimed → Assigned → Picked Up → Delivered → Verified

5. Main Features :

Surplus Food Listing : Donors can add food details, quantity, location, pickup deadline and contribution information.

Nearby Food : NGOs can browse available food and see important information before claiming it.

Urgency : Pickup deadlines help highlight food that needs to be collected quickly.

Volunteer Delivery : Volunteers can accept delivery requests and update the delivery status.

NGO Review : The NGO can confirm the delivery and rate quality, quantity and timeliness.

Impact Points : Successful donations and deliveries contribute to community scores.

Leaderboard : The leaderboard shows high-impact donors and volunteers.

Recognition : As a future feature, high-impact contributors can receive platform-funded media recognition.

6. User Interface :

The interface was designed to keep the main actions easy to find. The landing page gives the user three clear choices:

Food Donor, NGO / Recipient and Volunteer.

Donor Dashboard : The donor can see their impact, active donations and add new surplus food.

NGO Dashboard : The NGO can see available food nearby, claim food and track incoming deliveries.

Volunteer Dashboard : The volunteer can view delivery requests, accept them and update their delivery progress.

Leaderboard : The leaderboard gives a simple view of community contribution and helps make the impact visible.

7. Technology Used :

The prototype was intentionally kept simple because the hackathon focuses on solving the problem with minimal technology.

Frontend: HTML, CSS and JavaScript

Data handling: JavaScript application state

Persistence: Browser localStorage

Architecture: Single-page frontend

application with different role-based views The current prototype does not require a separate backend or database.

Local storage is used so that the important demo state can remain available after a page refresh.

8. Impact and Innovation :

The main goal of FoodResQ is to reduce the coordination gap between surplus food and people or organizations that can use it.

Expected impact :

- More surplus food can be redirected instead of being wasted.
- NGOs can discover available food more easily.
- Volunteers can help solve the last-mile delivery problem.
- Successful deliveries can be measured and verified.
- Community recognition can encourage people to contribute repeatedly.

What makes our approach different Instead of only creating a food listing system, FoodResQ connects the entire process.

The donor provides the food, the NGO claims it, the volunteer handles the delivery, and the NGO verifies the result.

This creates a simple loop:

contribution → delivery → verification → impact score → recognition.

9. Limitations and Future Scope :

Current limitations :

- The prototype uses browser storage instead of a centralized database.
- There is no real authentication or organization verification yet.
- Location and distance information is currently demo data.
- Real payment processing is not implemented. • Food safety verification is outside the current prototype.
- The media recognition idea is a future feature and is not based on existing newspaper partnerships.

Future improvements :

A future version could add a backend and database so multiple real users can use the platform at the same time. Real location services could be used to match food with nearby NGOs and volunteers.

Notifications and live delivery tracking could improve coordination. Payment support could be added for delivery costs or optional donor contributions. CSR partnerships and platform-funded recognition could be introduced to encourage long-term participation.

10. Conclusion :

FoodResQ is a simple prototype designed to make surplus food redistribution easier to coordinate.

It connects donors, NGOs and volunteers in one workflow and adds verification and community impact scores to encourage reliable participation.

The current version focuses on demonstrating the core idea clearly rather than trying to build every production feature.

With a backend, real location data and verified organizations, the same concept could be expanded into a larger food rescue network.

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GitHub:

<https://github.com/kartik-ey17/food-resq>

Demo:

<https://foodresq-xi.vercel.app>